

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The UDP protocol reveals that: The computer sent DNS requests to the DNS server using UDP

This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message: udp port 53 unreachable

The port noted in the error message is used for: DNS services

The most likely issue is: The DNS server was not reachable on UDP port 53

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

Time incident occurred: 13:24:32.192571

Explain how the IT team became aware of the incident: The IT team identified the issue by analyzing network traffic with the tcpdump network protocol analyzer

Explain the actions taken by the IT department to investigate the incident: The IT department captured and analyzed network packets to identify the protocols, IP addresses, ports, and error messages involved in the failed DNS requests

Note key findings of the IT department's investigation: The computer at 192.51.100.15 sent DNS requests using UDP to the DNS server at 203.0.113.2 on port 53. The requests failed repeatedly, and ICMP reported that UDP port 53 was unreachable

Note a likely cause of the incident: A likely cause is that the DNS service was unavailable or was not listening on UDP port 53. A firewall or incorrect network configuration could also have prevented access to the port

